

CACS101: Computer Fundamentals and Applications

Unit 2 – Computer Software

PRAJWAL BISTA

Introduction to Software

- Computer software, or just software, is a collection of computer programs and related data that provides the instructions for telling a computer what to do and how to do it.
- Any set of instructions that guides the hardware and tells it how to accomplish each task.
- Firmware: Firmware are also software (program or data) that has been permanently written onto ROM.

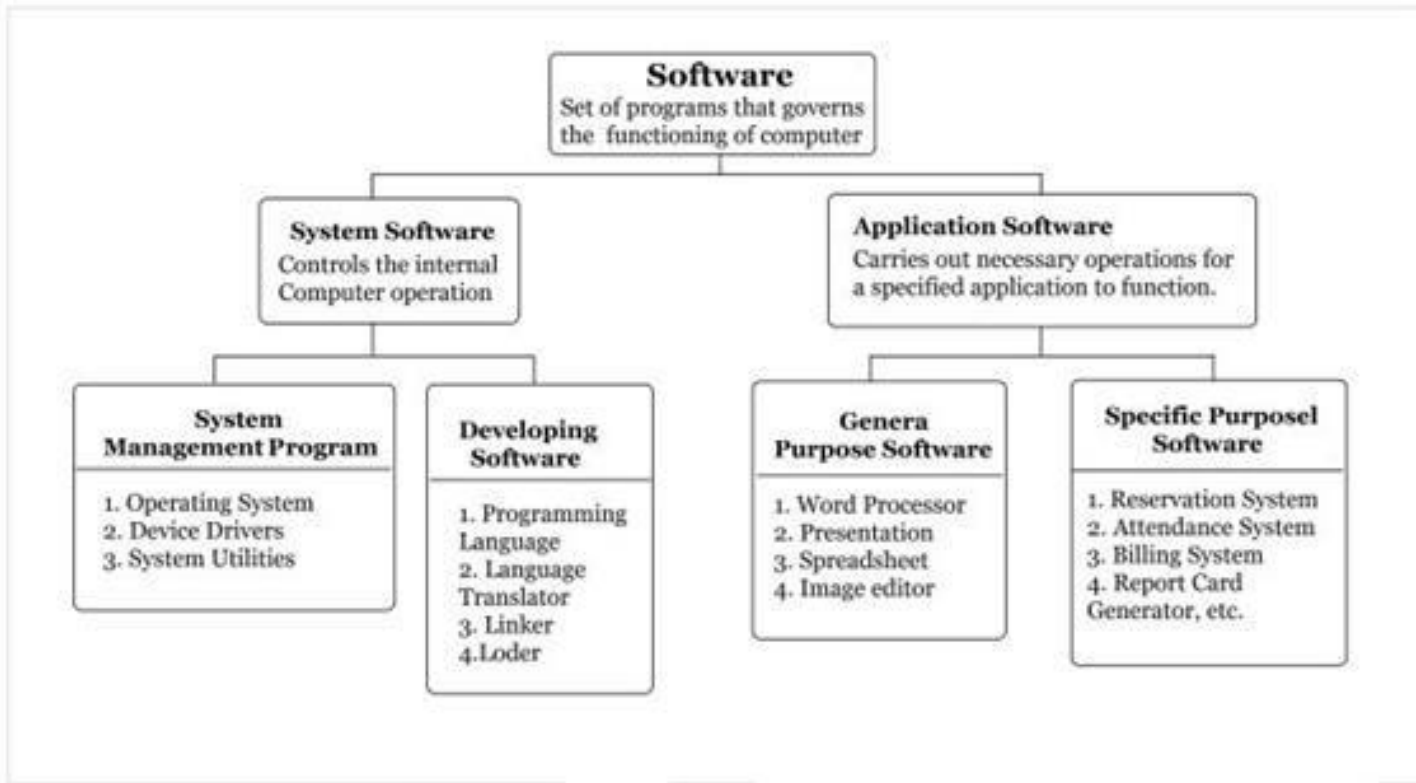
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- Software refers to the computer programs that are loaded into a computer system, and hardware refers to all the visible devices, which are assembled together to build a computer system. The blending of software and hardware gives life to a computer system.
- Hardware is of no use without software and software cannot be used without hardware.

Types of Software

- System Software
- Application Software
- Utility Software

Types of Software



System Software

- System software is computer software designed to operate the computer hardware to provide basic functionality and to provide a platform for running application software.
- It also provides the interface between the user and component of the computer.
- It directs the internal operation of computer system such as controlling I/O devices, computer hardware and managing the memory.

Utility Software

- Utility software is part of the system software and performs specific tasks to keep the computer running.
- Utility software is always running in the background.
- Examples of utility software are security and optimization programs. Security programs include anti-virus software that scans and removes viruses. Most computers will include some sort of anti-virus software, but you can add your own. Optimization programs can include tools for system clean-up, disk defragmentation, and file compression.
- These tools are typically installed as part of the operating system. They have access to the hard drive to keep it tidy.

Application Software

- The software that a user uses for accomplishing a specific task is the application software.
- Application software may be a single program or a set of programs. A set of programs that are written for a specific purpose and provide the required functionality is called software package.

System Software vs Application Software

System Software	Application Software
System Software maintain the system resources and give the path for application software to run.	Application software is built for specific tasks.
Its a general purpose software.	While its a specific purpose software.
Without system software, system can't run.	While without application software system always runs.
System software runs when system is turned on and stop when system is turned off.	While application software runs as per the user's request.
Example of system software are operating system, etc.	Example of application software are Photoshop, VLC player etc.

Program vs Software

Program	Software
Program is a set of instructions written in a programming language used to execute for a specific task or particular function.	Software is a set of Programs used to execute for an entire application. There are Many programs combine together to form software. It is also used to perform a task.
A program does not have further categorization.	Software can be categorized into two categories: application software and system software.
Program size ranges in between kilobytes and megabytes.	Software size can range from megabytes to gigabytes.
Dependent on Compiler	Dependent on operating System

Program vs Software

Program	Software
A program is compiled every time when we need to generate some output from it.	Whole software is compiled, tested and debugged during the development process.
Program has limited functionality and less features.	Software has lots of functionality and features such as GUI, input/output data, process etc.
It has no documentations only comments	It has documentations
E.g. add two numbers, factorial, greatest of two number, greatest of three number,etc.	E.g. Application software are: Microsoft Word, Microsoft Excel, Google Chrome, VLC media player, Firefox, Adobe Reader etc. System Software are: Windows, Linux, Unix, Mac etc.

Computer Virus

- Computer Virus is a software program that is destructive in nature.
- It can attach itself to other healthy programs.
- It can replicate itself and thus can spread across a network.
- It is difficult to trace a virus after it has spread across a network.
- Viruses harm the computer in many ways
 - corrupt or delete data or files on the computer,
 - change the functionality of software applications,
 - use email program to spread itself to other computers,
 - erase everything on the hard disk, or,
 - degrade performance of the system by utilizing resources such as memory or disk space.

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- Virus infects an executable file or program. The virus executes when a program infected with virus is executed or you start a computer from a disk that has infected system files.
- Once a virus is active, it loads into the computer's memory and may save itself to the hard drive or copies itself to applications or system files on the disk.
- Some examples of viruses are: "Melissa" and "I Love You"

Signs of Virus Infection

- Slower system performance
- Pop-ups bombarding the screen
- Programs running on their own
- Files multiplying/duplicating on their own
- New files or programs in the computer
- Files, folders or programs getting deleted or corrupted

Antivirus

- Antivirus is a type of program designed and developed to protect a computer from malware like computer virus, worm, spyware, botnets, boot-kits, keylogger, etc.
- Antivirus function to scan, detect and remove such viruses from the computer.
- Most antivirus incorporates both automated and manual filtering abilities. Instant scanning option may check files downloaded from the internet, disks that are embedded into PCs and files that are made by software installers.
- Example: Avast, Kaspersky, etc.